Reading a Month of Attack Traffic: A Power BI Analysis of 148,580 Honeypot Connections

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Data*: one T-Pot/Cowrie honeypot sensor, Sept 3 to Oct 3, 2026 (UTC). All IP addresses in this report are masked to protect the owners of those addresses.*

At a glance: Built in Power BI (Power Query, DAX, data modeling) from 31 daily log files: about 483,000 events, 148,580 connections and 7,103 distinct IPs. Deliverables: a four-page dashboard and this report.

Summary

This report analyzes 31 days of connection logs from a single honeypot, a decoy server that accepts SSH and Telnet logins so that automated attackers reveal what they try. I turned roughly 483,000 log events into a four-page Power BI dashboard built around four questions: when activity happens, where it comes from, how sources compare with each other, and what attackers do once they get in. I did not set out to judge the sensor or its security. The aim was to find meaningful trends and to show how I got to them.

The main findings:

- **Volume is driven by a few sources, not by many.** The sensor logged 148,580 connections from 7,103 distinct IP addresses. The largest daily spike (Sept 6, 14,258 connections) came mostly from a single address that ran at a steady rate for about four hours. Other busy days look the opposite: hundreds of distinct IPs, each making only a few connections.
- **Two address blocks account for 38% of all connections.** One block of 256 addresses (a /24) located in Bulgaria produced 26.2% of all connections from 79 IPs, and a block located in the US produced 11.8% from 43 IPs. Together, 122 IPs (1.7% of sources) made 38% of the traffic.
- **Few sessions get in, and those that do follow a script.** About 77% of connections try a login and 4.3% of those succeed. Of the 4,901 successful logins, 72% run commands Only 61 distinct files were ever captured, and one of them accounts for 79% of file-capture sessions.
- **Usernames are concentrated, passwords are not.** The username 'root' alone is 38% of login attempts, while the ten most common passwords together are only 16%.

- **Protocol choice varies sharply by country.** Traffic from Pakistan is 92% Telnet, while traffic from Bulgaria, Poland and Romania is more than 99.7% SSH.

The repeated lesson I learned from this was simple: headline totals and averages repeatedly hid the real structure. Actually analyzing the distributions, and comparing volume next to the number of distinct sources, changed the interpretation almost every time.

1. Background and Goal

SImply put, a honeypot server is a deliberately vulnerable system that is intended to be attacked, for the sole purpose of safely overseeing these attackers and collecting data on them.. T-Pot is a platform that bundles many honeypots together for rental. I worked with Cowrie server, which imitates SSH and Telnet servers: it accepts logins, records every command typed, and captures any files that attackers send or fetch.

A friend who rented the servers shared the logs with me. He wrote his own threat-research report from the same data, so this report is independent and has a different purpose: it is analytical, aimed at trends and at the data-handling work needed to find them. I chose to build the dashboard and process the data in Power BI because that was a skills gap I wanted to close.

My goal was to show that I can take a large, messy, real-world dataset from raw files to a tested dashboard and a written analysis, all while documenting my decisions, questions and mistakes along the way. I organized the work around four angles: time series (when), geography (where from), comparison (who, and how they differ) and post-compromise behavior (what happens after a login).

2. The Data

- **Format:** 31 daily files in JSON Lines, compressed with gzip apart from the newest day. The logs are an event stream: each connection produces a sequence of events (connect, login attempts, commands, close), and the fields differ by event type.
- **Window:** Sept 3 00:00 to Oct 3 10:00 UTC. The first day is complete; the last is a partial day.
- **Scope:** the T-Pot export contained many honeypot types like Dionaea, Tanner, adbhoney, and spiderfoot just to name a few. I used the Cowrie log only because it was one of the richest and most interpretable, and is a very common honeypot for data analysts to work with.
- **Integrity and safety:** I verified the download against its SHA-256 checksum. The files were treated as potentially hostile: nothing executable was run, no links were opened, and captured files were analyzed only by hash and count.
- **Privacy:** To avoid doxxing anyone, single IPs show the last number as 'xxx', address ranges appear as /24 blocks, and the dashboard's only IP-level table is masked. The raw logs are also not shared or public.

3. Preparing the Data

I combined the 31 files for the 31 days in Power Query, filtered to the events I needed, and kept every transformation as a documented step. While combining the data was relatively easy, these problems were the true work:

- **Data loading that hung for hours.** One of if not the biggest difficulties was actually the time spent waiting for queries to finish or data to be loaded. The large quantity of data being used combined with the less than optimal laptop are the main culprits. SSH client-fingerprint events also carry large nested arrays that Power Query parsed before filtering. Additionally, every queries ran up stream through every single event, which was in the hundreds of thousands. This resulted in a very warm, though determined, laptop, and a very slow computing time.
- **Timestamps shifted six hours.** A partial Sept 2 appeared even though collection starts Sept 3 UTC. A check on the earliest timestamp confirmed Power Query had converted UTC to local time, so I added a UTC column and rebuilt the date from it.
- **Geolocation needed a range match.** The country lookup maps blocks of addresses, so a plain join fails. I converted addresses to numbers, geolocated only the distinct IPs (thousands of rows, not hundreds of thousands to avoid that wait time), and linked the result to the events with a relationship, because a merge made Power Query re-run the whole pipeline on every preview again causing more processing and waiting.
- **Shortcuts and averages gave wrong or misleading numbers.** This happened many times where the data would either default to a 'percent of grand total' divided by the visible rows only. This skewed the findings significantly, but was a simple fix.

  I also cut down the total number of columns. When first imported there were 29 columns. I ended up keeping 15, deciding only on keeping a column if it fed one of the four questions. Each removal has a documented reason.

4. Dashboard Design

The dashboard has four pages, one per question: Time Series, Geolocation, Comparative Analysis and Post-Compromise. These were the basic principles behind each dashboard to ensure consistency and quality:

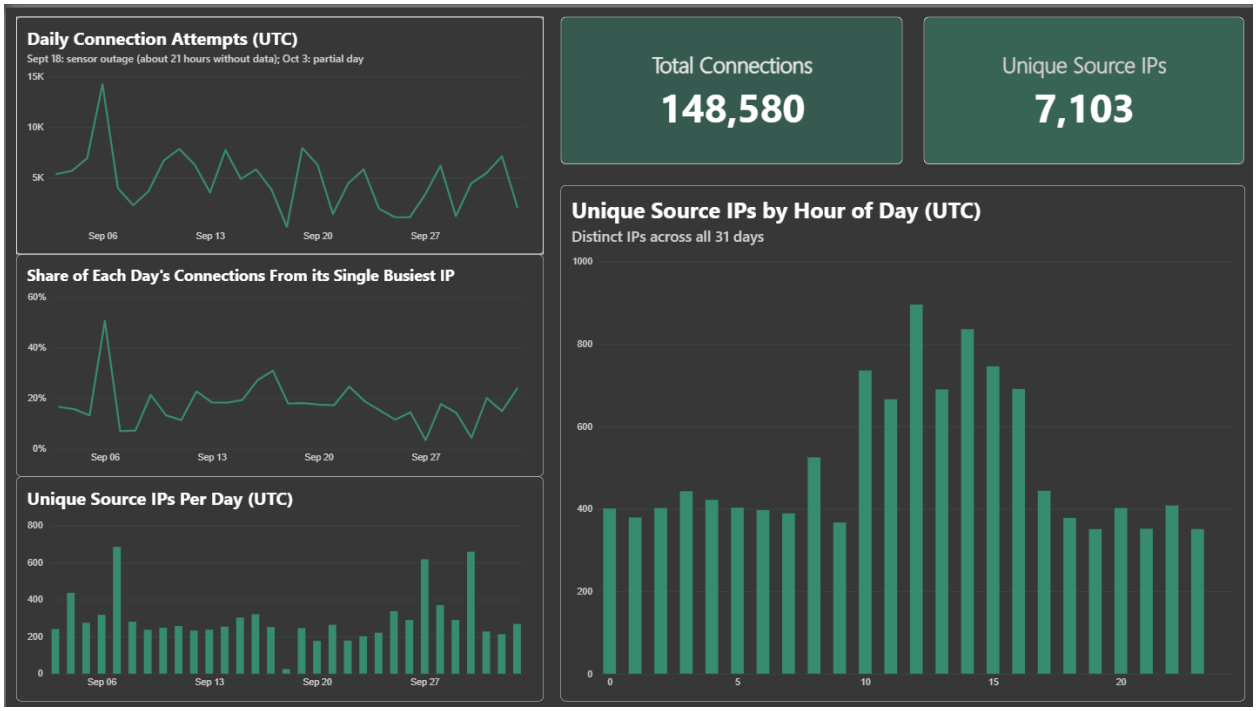
- **Volume is always shown next to breadth (distinct IPs)**, one heavy IP cannot pass for a trend.
- **The same color always means the same measure,** and every title says what is measured and in which time zone.
- **Every number reconciles with another view:** funnel stages equal the card values, and shares add up to their totals.
- **Investigation views stay off the dashboard.** I had an exploratory page cluttered with tables, more so a hidden scratch page, with the findings written up here. It mainly

went into more depth on the Sept 6th events, analysis on Poland and Belgium, the sensor outage on the 18th, and few scratch tables for reference.

The model is an events table plus a small distinct-IP table linked by a relationship, with all calculations written as measures in DAX and explained in the Appendix.

5. Findings

5.1 When: There are Two Kinds of Spikes

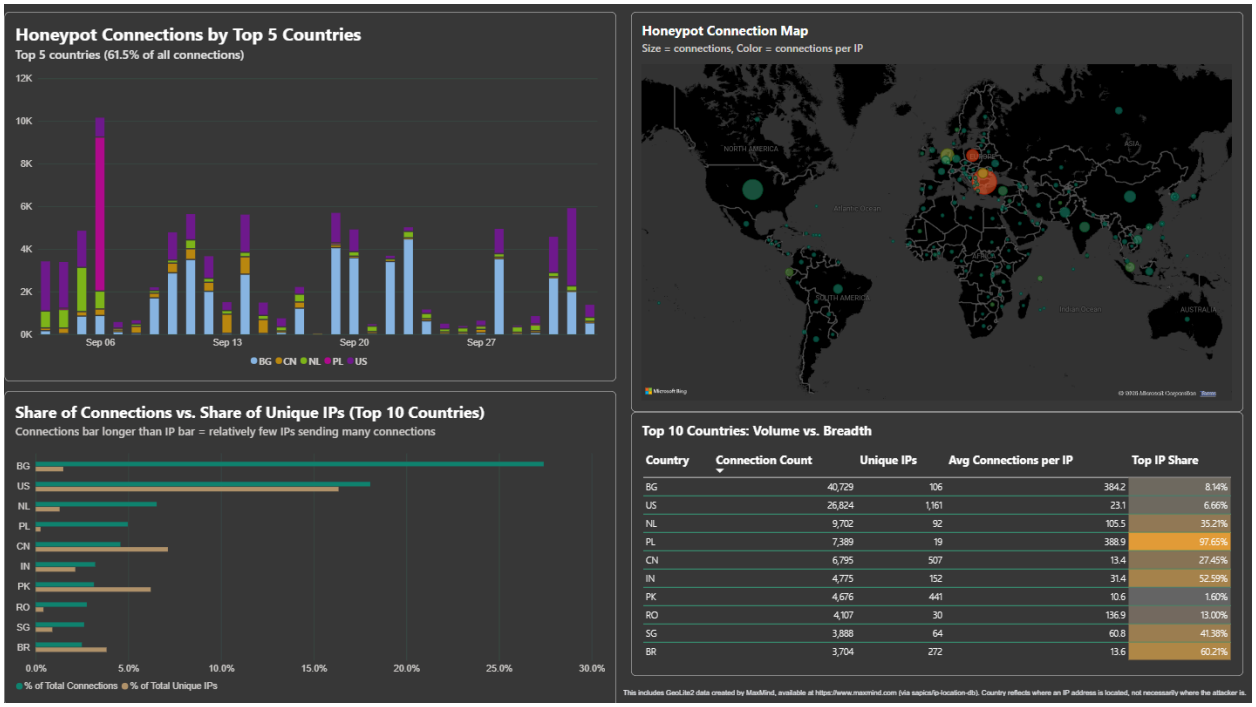


Most days recorded between 1,000 and 8,000 connection attempts, with an average of approximately 4,800 per day. The largest spike occurred on September 6, when the server received 14,258 connections from 318 unique source IPs. Notably, a single IP address from Poland accounted for 7,215 of those connections, representing 51% of the day's total activity. This address did not appear on any other day in the dataset. Its activity was concentrated within roughly four hours and followed a nearly constant rate of approximately 1,800 connections per hour, or about one connection every two seconds. The consistency and concentration of this activity are characteristic of an automated process rather than normal, irregular user activity.

The number of unique source IPs per day reveals a different pattern. On most days, approximately 250–350 distinct IP addresses connected to the server, while several days showed substantially higher counts. September 7 recorded 685 unique IPs despite generating only about 4,000 total connections. This suggests two distinct types of activity spikes: bursts, where a small number of IP addresses generate a large volume of connections, and sweeps, where a much larger number of IP addresses make relatively few connections each.

The daily share of connections generated by the single busiest IP helps distinguish between these two patterns. Days with a high concentration from one IP are more consistent with burst activity, while days with many unique IPs and lower individual contribution are more indicative of a sweep. The hourly distribution of unique source IPs also shows a clear time-of-day pattern. Distinct IP activity more than doubles between approximately 10:00 and 16:00 UTC, suggesting that the server experienced substantially broader scanning or connection activity during these hours.

5.2 Where From: A Few Address Blocks Dominate



The ten countries generating the most connections account for 75.8% of all connections but only 40% of distinct source IPs, showing that connection volume is highly concentrated among a relatively small portion of the global source population. Bulgaria alone generated 40,729 connections, or 27.4% of all activity, from only 106 unique IPs, representing approximately 1.5% of all source addresses. The United States generated the second-highest volume, with 26,824 connections from 1,161 unique IPs. This difference really shows why connection totals alone do not fully describe the nature of activity originating from a country.

Average connections per IP can also be misleading when activity is concentrated among a small number of sources. Poland, for example, averaged 388.9 connections per IP, but 98% of the country's traffic came from a single address. Brazil and India show a similar pattern, with 60% and 53% of their respective traffic coming from a single IP. Bulgaria presents a very different pattern: its busiest IP accounted for only 8% of the country's connections, indicating that its high volume was distributed across many sources rather than driven by one dominant address.

The /24 block-level analysis provides additional context and reveals concentration that is not immediately apparent at the country level. One /24 block in Bulgaria generated 38,949 connections, approximately 96% of Bulgaria's total traffic, across 79 unique IPs. Similarly, a U.S. /24 block generated 17,537 connections from 43 IPs, accounting for roughly two-thirds of all U.S. activity. This changed the initial interpretation of the U.S. traffic: what appeared at the country level to be broad background activity was, at the network-block level, much more concentrated.

Taken together, the sources fall into three distinct patterns. Always-on fleets generate sustained activity from a concentrated network range, as seen in the Bulgarian and U.S. blocks. Single-host bursts are characterized by one IP generating a disproportionate share of a country's traffic, as seen with the Polish source. Finally, wide, shallow scanning involves many IPs making relatively few connections each, producing broad geographic coverage without a dominant source. Pakistan and China provide examples of this third pattern. Distinguishing these patterns is more informative than ranking countries by connection volume alone because it reveals how the observed activity is actually distributed across individual sources and network ranges.

5.3 How sources compare: credentials, protocols and top IPs



Credentials

The attempted usernames show a strong concentration around common administrative accounts. The username “root” appeared in 38.4% of approximately 123,000 login attempts,

representing 47,179 attempts from 1,711 unique IP addresses. The ten most frequently attempted usernames together accounted for approximately 55% of all username attempts. Password attempts were considerably more distributed: the ten most common passwords accounted for only about 16% of all passwords attempted, led by 123456, 123, and 1234. Each of these passwords was attempted by roughly 190–490 distinct IP addresses, generally with only a small number of attempts per IP. This distribution is more consistent with broadly shared credential lists being used across many sources than with a single operator repeatedly testing its own password set.

Protocols

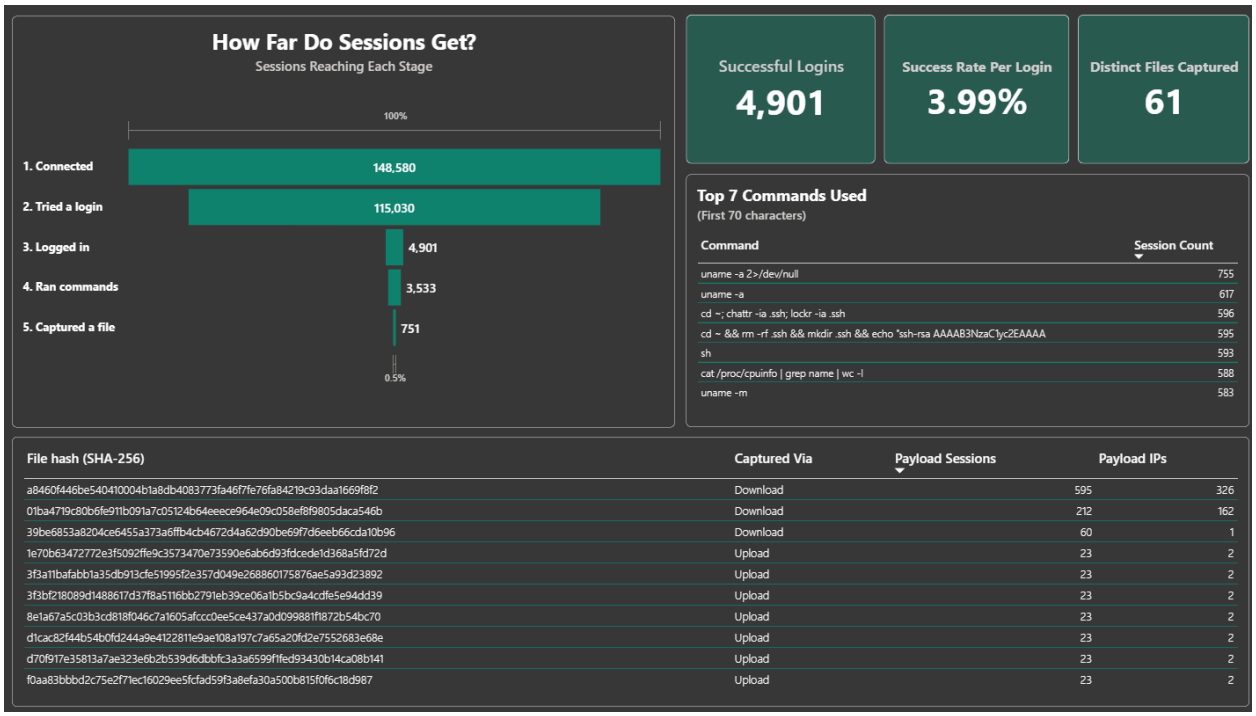
Only two exposed services were targeted in the dataset: SSH on port 22 and Telnet on port 23. The balance between these protocols varies substantially by country. Pakistan's activity was 92% Telnet, spread across 441 unique IP addresses with no single dominant source. Belgium showed a similar pattern, with 93% of its connections targeting Telnet. In contrast, activity from Bulgaria, Poland, and Romania was more than 99.7% SSH. This suggests that protocol preference is not uniform across the source population and may reflect differences in the tools, scanning behavior, or infrastructure used by different groups of sources.

Top Source IPs

The ten busiest source IPs accounted for 20.4% of all connections, but their activity patterns were not uniform. Some sources demonstrated persistent, low-volume activity, while others appeared only briefly but generated large numbers of connections. A host in the Netherlands was active on all 31 days, averaging approximately 110 connections per day. Sources in India and Turkey were active on 19 and 16 days, respectively, showing similarly sustained activity. In contrast, the Polish source appeared on only one day, while three addresses within the Bulgarian network block were active for just two to four days each but generated approximately 630–1,200 connections per day.

These differences reinforce the broader patterns identified throughout the analysis. Some sources behave like persistent background scanners, maintaining activity over an extended period, while others exhibit short, high-volume bursts before disappearing. Looking only at total connection counts would group these sources together, but examining activity duration and connections per day reveals substantially different behaviors.

5.4 What happens after login



The commands executed after successful login suggest a sequence of system reconnaissance followed by a possible SSH-key persistence routine. Several commands appear to inspect the environment before removing the existing .ssh directory and installing a key controlled by the source. Five commands each appeared in approximately 583–596 sessions, a level of consistency that suggests they may be components of a scripted sequence executed from start to finish. If confirmed, the SSH-key activity would indicate an attempt to maintain access beyond the initial login.

File-capture activity was comparatively limited and concentrated among a small number of recurring files. Across the 751 file-capture sessions, only 61 distinct files were recorded: 21 downloaded and 40 uploaded. The most frequently observed file appeared in 595 sessions, accounting for approximately 79% of all file-capture sessions. Its frequency suggests a possible relationship with the SSH-key installation activity, although this connection has not yet been verified at the individual-session level.

A second pattern emerged among the uploaded files. Seven files each appeared in exactly 23 sessions associated with the same two source IPs. This repeated distribution is consistent with the possibility that the files were uploaded together as part of a shared toolkit or scripted operation. However, the session-level relationships have not been confirmed. Both the potential link between file capture and key installation and the possibility of bundled file uploads should therefore be treated as working hypotheses rather than established findings.

Overall, the results show a clear progression from widespread login activity to a smaller set of sessions involving command executions and file transfers. The repeated command patterns and

concentration of file captures suggest some degree of automation, while the unverified relationships between commands, files, and source IPs identify useful targets for potential further investigation.

6. Putting It Together

The observed traffic falls into three broad patterns: always-on fleets operating across network blocks over multiple days, single-host bursts lasting hours or a single day, and wide, shallow background scanning from thousands of IPs. Raw connection counts blend these behaviors together, making it difficult to distinguish volume from persistence or source diversity. Comparing connection totals with unique IPs and analyzing activity at both the IP and network-block levels reveals a much clearer picture.

The attack funnel reinforces this pattern: although credential guessing occurs at scale, a much smaller share of sessions results in successful logins, command execution, and file capture. Together, these findings show why understanding honeypot activity requires looking beyond connection volume to examine how sources behave and how far they progress after gaining access.

7. Limitations

- **Limited scope.** The analysis covers one honeypot sensor over one month, so the observed patterns may reflect this sensor's configuration and exposure rather than broader internet activity. The 4% login success rate describes this honeypot, not real-world servers.
- **Approximate geolocation.** Country-level locations indicate where IP addresses are registered or geolocated, not necessarily where attackers are physically located. Network blocks may also contain addresses used by different customers, so a shared block does not necessarily represent a single actor. The map provides approximate country-level locations, not exact physical positions.
- **File counts do not equal malware counts.** The 61 distinct files captured represent unique files, not necessarily 61 different types of malware.
- **Incomplete data.** A sensor outage of approximately 21 hours on September 18 and partial data for October 3 may have caused activity on those dates to be undercounted.

8. What I Learned

- **Check the denominator and distribution.** Totals, averages, and built-in percentages repeatedly created misleading initial impressions. Rechecking the underlying data changed my interpretation and, in several cases, the conclusions themselves.
- **Validate fields before using them.** The file-hash column contained two different types of values. Identifying this issue prevented the payload count from being inflated by approximately 28 times.
- **Design the data model to avoid rework.** Restructuring a merge into a relationship and filtering records before parsing eliminated repeated processing steps that had taken roughly an hour each.
- **Document decisions throughout the process.** Maintaining a running log of questions, changes, and corrections made it possible to trace errors, verify assumptions, and resolve issues as the analysis progressed.

Appendix: Key Measures

- **Connection Count:** connection events (one per session).
- **Unique Source IPs:** distinct IPs that connected.
- **Connections per IP:** connections divided by unique IPs; always read together with Top IP Share.
- **Top IP Share:** the share of a group's connections made by its single busiest IP.
- **Share of Connections / Share of Unique IPs:** a group's share of all connections and of all distinct IPs.
- **Login Attempts / Successful Logins:** login events, failed plus successful, and successful only.
- **Active Days:** distinct UTC days on which an IP made at least one connection.
- **Sessions per funnel stage:** distinct sessions that connected, tried a login, logged in, ran commands, and captured a file.
- **Distinct Files Captured:** distinct file hashes from download and upload events only.
- **Subnet (/24) and IP Label:** the first three numbers of an address plus '.0/24', and a masked label that keeps the last number hidden.